Khulna University of Engineering & Technology (KUET)**Khulna University of Engineering & Technology (KUET)**

Department of Computer Science and Engineering

CSE 4112: Machine Learning Laboratory

Project Proposal

University Student Mental Stress Pattern Analysis Using Unsupervised Learning

A Study of Academic Stress Patterns Among KUET Engineering Students Using Clustering

Course	CSE 4111 / CSE 4112 – Machine Learning / Machine Learning Laboratory
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Group Members	[Name 1 – Roll] [Name 2 – Roll] [Name 3 – Roll]
Target Population	KUET CSE undergraduate students (1st–4th year), extendable to other departments
Date	[Submission Date]

1. Introduction

Academic stress is a well-documented factor affecting the mental health, performance, and retention of engineering students worldwide, and Bangladeshi engineering students face a distinctive combination of pressures — rigorous coursework and CGPA-driven competition, limited career and financial security, and social or family expectations. Despite this, systematic, data-driven studies that quantify and segment stress patterns within a single Bangladeshi engineering institution are scarce. Most existing work either treats stress as a single predicted label (e.g., "high" vs. "low") or relies on qualitative interviews with small sample sizes.

This project proposes an unsupervised, data-driven approach: rather than forcing students into pre-defined stress categories, we let natural groupings emerge directly from their survey responses. This is more appropriate for a first exploratory study, since there is no verified ground-truth label for "stress level" and the goal is to discover what distinct stress profiles actually exist among KUET students, before any predictive model is built on top of them.

2. Problem Statement

Given anonymous survey responses from KUET engineering students covering academic workload, exam anxiety, motivation, peer comparison, career/financial pressure, and support-seeking behavior, can we use unsupervised machine learning to:

- Identify distinct, interpretable clusters of students who share similar stress profiles;
- Determine which features (e.g., sleep sacrifice, backlog courses, CGPA comparison, financial worry) most strongly differentiate these clusters; and
- Translate these clusters into actionable insight for students and university counseling/academic support services?

3. Objectives

- Design and deploy an anonymous, validated survey instrument targeting KUET CSE students across all four academic years (and optionally other departments).
- Build a clean, well-documented dataset of student stress-related responses suitable for unsupervised analysis.
- Apply dimensionality reduction (PCA) to manage the feature space and visualize student distribution in reduced dimensions.
- Apply K-Means and Hierarchical Agglomerative Clustering to segment students into stress-pattern groups, and determine the most defensible number of clusters using internal validation metrics.
- Profile and interpret each cluster in plain language (e.g., "High-Achieving but Isolated," "Financially Burdened," "Resilient/Well-Supported") and connect findings to concrete recommendations.

4. Data in Brief

As a quality-check reference for our survey design, we consulted a 2020 Data in Brief article reporting a survey-based dataset on university student wellbeing and stress-related indicators (ScienceDirect: https://www.sciencedirect.com/science/article/pii/S2352340920312087). Its variable structure — combining Likert-scale psychological items with demographic and lifestyle variables — informed the structure of our own questionnaire, though our instrument is independently designed and localized to the KUET engineering context.

5. Target Population and Sampling Strategy

The primary target population is KUET CSE undergraduate students across four academic years, with approximately 120 students per batch (≈480 students total). If time and access permit, the survey can be extended to other KUET departments to test whether stress patterns are CSE-specific or generalize across engineering disciplines. Distribution will be stratified by year, since stress drivers plausibly differ across years (e.g., 1st years adjusting to university life vs. 4th years facing job-market anxiety).

6. Data Collection: Survey Instrument

Data will be collected via an anonymous Google Form (estimated completion time: 4–5 minutes), explicitly stating that participation is voluntary, responses are confidential, and there are no right or wrong answers. The instrument has three parts:

6.1 Part 1 – Demographic and Contextual Variables

#	Item	Response Options
1	Current Year	1st / 2nd / 3rd / 4th Year / Other
2	Current CGPA	Below 2.50, 2.50–2.99, 3.00–3.49, 3.50–3.79, 3.80–4.00
3	Gender	Male / Female
4	Living Arrangement	Hall / Family / Mess / Other

6.2 Part 2 – Stress Experience (5-point Likert, 1 = Strongly Disagree to 5 = Strongly Agree, unless noted)

#	Statement
5	I sacrifice sleep to keep up with coursework.
6	Assignments and lab reports pile up faster than I can finish them.
7	I worry about failing exams even when I have prepared well.
8	One poor result hurts my motivation for the rest of the semester.
9	I have one or more backlog/incomplete courses. (Yes / No)
10	I constantly compare my CGPA with my classmates.
11	I worry about landing a good engineering job after graduation.
12	I feel comfortable asking teachers for academic help.
13	Feedback from instructors actually reduces my stress.
14	Financial concerns interfere with my academic performance.
15	I feel pressure to start earning as soon as possible.

6.3 Part 3 – Open-Ended

16. “In one or two sentences, what is your single biggest source of academic stress?” — This free-text item is not used directly in the clustering step, but will optionally support a light qualitative/NLP pass (e.g., keyword or theme frequency per cluster) to sanity-check and enrich the quantitative cluster profiles.

7. Data Preprocessing Plan

- Remove incomplete/duplicate submissions and validate response ranges.
- Encode ordinal demographic fields (Year, CGPA band) as ordinal integers; one-hot encode nominal fields (Gender, Living Arrangement); binary-encode the backlog Yes/No item.
- Keep the 10 Likert items (Q5–Q8, Q10–Q15) as numeric features on their native 1–5 scale.
- Standardize all numeric features (zero mean, unit variance) before clustering, since K-Means and PCA are both distance/variance-sensitive.
- Check for and document missing values, response bias (e.g., straight-lining), and class imbalance (e.g., gender skew, which is common in CSE cohorts) as part of the dataset documentation sheet required by the course.

8. Proposed Machine Learning Methodology

ML Task Type: Unsupervised Learning — Clustering, combined with Dimensionality Reduction (PCA) for visualization and to mitigate the curse of dimensionality across the encoded feature set.

8.1 Dimensionality Reduction

Principal Component Analysis (PCA) will be applied to the standardized feature matrix to (a) visualize students in 2D/3D space before committing to a clustering algorithm, (b) reduce noise/redundancy among correlated Likert items (e.g., sleep sacrifice and assignment overload are likely correlated), and (c) report explained variance per component so the reduction is justified rather than arbitrary.

8.2 Clustering Algorithms

Two complementary clustering methods will be used, both covered in the CSE 4111 syllabus and both well suited to this dataset:

- K-Means Partitional Clustering — the primary method, chosen for its scalability to ~250+ responses and ease of interpretation. The optimal k will be chosen using the Elbow Method (within-cluster sum of squares) combined with the Silhouette Score, rather than an arbitrary guess.
- Hierarchical Agglomerative Clustering — used as a cross-check via dendrogram inspection, which is especially useful here because it does not require pre-specifying k and can reveal whether clusters are nested (e.g., a broad “high-stress” group that splits into “financially driven” and “academically driven” sub-groups).

Because the dataset mixes continuous Likert items with categorical demographics, we will primarily cluster on the standardized numeric/encoded feature set; as a documented limitation and possible extension, we note that a mixed-data method such as K-Prototypes or Gower-distance-based clustering could better respect the categorical variables if time permits.

8.3 Cluster Validation

- Silhouette Score and Davies–Bouldin Index to quantitatively compare candidate values of k.
- Visual inspection of PCA-projected clusters for separation and overlap.
- Stability check: re-run clustering on bootstrap sub-samples to confirm clusters are not an artifact of one sample draw.

9. Cluster Profiling and Interpretation

Once clusters are finalized, each will be profiled using the mean/mode of its member features (e.g., average sleep-sacrifice score, backlog prevalence, CGPA band distribution, financial-concern score) to produce a human-readable persona per cluster — for example, a cluster with high exam anxiety, high CGPA comparison, but low backlog rate might be labeled "High-Achieving but Anxious," while a cluster with high financial-concern and high sleep-sacrifice scores might be labeled "Financially Strained." Where useful, keyword frequency from the open-ended Q16 responses within each cluster will be used to qualitatively support these labels.

10. Expected Outcomes and Benefits

10.1 For Students

- Increased awareness that stress is patterned and shared, not an individual failing — anonymized, aggregate results can be shared back with participants.
- A basis for peer-support or study groups organized around shared stress profiles (e.g., students facing similar exam-anxiety patterns).

10.2 For the Institution / Department

- Evidence-based input for the counseling office and academic advising to design targeted interventions per cluster rather than one-size-fits-all wellness programs.
- Early-warning signal: if a cluster combining high backlog rate, high sleep-sacrifice, and low comfort seeking help is identified, the department can proactively target that group with academic support before performance deteriorates further.
- A reusable, low-cost survey-and-analysis pipeline the department can re-run each semester or year to track whether stress patterns shift (e.g., after curriculum or policy changes).

10.3 For the Course (CSE 4112)

- A complete, locally relevant, Bangladesh-specific unsupervised learning case study covering the full pipeline: survey design → preprocessing → PCA → clustering → validation → interpretation — satisfying the Data in Brief and Bangladesh-context requirements of the lab.

11. Ethical Considerations

- Participation is voluntary and anonymous; no names, roll numbers, or other directly identifying information will be collected.
- The form explicitly states the purpose, confidentiality, and voluntary nature of participation before any question is asked.
- Aggregate/cluster-level results only will be reported — no individual response will be traceable or shared.
- Since the topic concerns mental stress, the form and any published results will avoid diagnostic or clinical language; students showing signs of significant distress in open-text responses will not be individually contacted, and the form will include a note pointing to KUET's counseling resources for anyone who wants support.
- Data will be stored securely and deleted after course evaluation unless explicit consent for longer retention is given.

12. Limitations

- Self-selection bias: students who are unusually stressed (or unusually relaxed) may be more/less likely to respond, skewing cluster proportions.
- Self-reported Likert data reflects perceived rather than clinically measured stress, and is subject to social-desirability and recall bias.
- K-Means assumes roughly spherical, similarly sized clusters and is sensitive to the mix of categorical and continuous features after encoding; results should be cross-checked against the hierarchical clustering output.

- Findings are correlational, not causal — clusters describe co-occurring patterns, not what causes stress.
- Results are specific to KUET (and primarily CSE) and may not generalize to other institutions or disciplines without re-validation.
- No exact Bangladesh-specific Data in Brief dataset on engineering-student stress was found; the closest available reference (a 2020 general university wellbeing survey dataset) was used only as a design reference, not as supplementary training data.

13. Work Plan / Timeline

Week(s)	Activity
1	Finalize survey instrument, get instructor sign-off, prepare dataset documentation sheet.
2–3	Distribute Google Form across all four CSE years (and other departments if extended); track response counts per stratum.
4	Clean and preprocess data; encode features; produce descriptive statistics.
5	Apply PCA; run K-Means (Elbow/Silhouette) and Hierarchical Clustering; validate cluster count.
6	Profile clusters, interpret results, draft recommendations.
7	Compile final report, visualizations, and presentation for submission.

14. Conclusion

This project applies unsupervised learning — PCA for dimensionality reduction together with K-Means and Hierarchical Agglomerative Clustering — to anonymous survey data from KUET engineering students, in order to surface distinct, interpretable stress-pattern groups rather than a single predicted stress score. Beyond fulfilling the CSE 4112 lab requirements, the pipeline is designed to be genuinely reusable by the department for ongoing, low-cost monitoring of student wellbeing, and to produce findings that can directly inform how academic and counseling support is targeted at KUET.

15. References

1. Data in Brief – University student wellbeing/stress-related survey dataset (2020)

2. student-stress-unsupervised-learning (GitHub reference codebase)

3. Academic Stress among Bangladeshi Engineering Students — primary survey instrument (Google Forms), designed for this project.